

Survey Data Analysis Report

Overview

Shape of the dataset: (112, 35)

Summary statistics:

What type of field do you work in? What is the type of organization your laboratory operates under? In which country is your laboratory or organization based ? How many years of experience do you have in your field? What is your role in the organization? How familiar are you with AI technologies in laboratory operations? To what extent is AI currently used in your laboratory operations? Which AI applications are used in your laboratory? (Select all that apply) To what extent do you agree with the following statement? [AI has improved operational efficiency in my laboratory.] To what extent do you agree with the following statement? [AI has enhanced diagnostic accuracy and consistency.] To what extent do you agree with the following statement? [AI has strengthened data management and security capabilities.] Rate the impact of AI on the following laboratory area: [Automation of routine tasks] Rate the impact of AI on the following laboratory area: [Data analysis and management] Rate the impact of AI on the following laboratory area: [Speed and accuracy of diagnostics] Rate the impact of AI on the following laboratory area: [Workflow optimization] To what extent do the following barriers affect AI implementation in your laboratory? [Financial constraints] To what extent do the following barriers affect AI implementation in your laboratory? [Ethical concerns (e.g., data privacy, transparency)] To what extent do the following barriers affect AI implementation in your laboratory? [Staff resistance to change] To what extent do the following barriers affect AI implementation in your laboratory? [Lack of training and education] To what extent do the following barriers affect AI implementation in your laboratory? [Regulatory compliance issues] Which strategies have been implemented in your laboratory to overcome AI challenges? (Select all that apply) How effective have the following strategies been in overcoming

challenges? [Staff training and upskilling] How effective have the following strategies been in overcoming challenges? [Vendor partnerships] How effective have the following strategies been in overcoming challenges? [Data management protocols] How effective have the following strategies been in overcoming challenges? [Scalable solutions] If you could prioritize one strategy, which would it be? How important are the following criteria when selecting AI implementation strategies?\n [Cost-efficiency] How important are the following criteria when selecting AI implementation strategies?\n [Scalability] How important are the following criteria when selecting AI implementation strategies?\n [Ethical compliance] How important are the following criteria when selecting AI implementation strategies?\n [Accuracy and reliability] How important are the following criteria when selecting AI implementation strategies?\n [Ease of integration] What factors influence your decision-making process when selecting an AI strategy? (Select all that apply) What do you perceive as the biggest benefits of AI implementation in your laboratory? What are the most significant challenges you've faced in implementing AI? How do you envision AI evolving in your laboratory over the next five years?

count	112	112
112	112	112
112	112	112
109		112
	112	
		112
	112	112
112		112
	112	
112		112
	112	
		112
112		112
	112	
		112

112

112

112

112

112

112

112

112

112

111

86

80

78

unique

35

2

16

4

21

4

4

24

5

5

5

5

5

5

5

5

5

5

5

5

19

5

5

5

5

4

5

5

5

5

5

16			84
77			76
top	Clinical Laboratory		Private/Commercial Organization
	Saudi Arabia	More than 5 years	Laboratory
Technician or Specialist		Familiar	
Limited use		Data management	
	Agree		
	Agree		
Agree		Moderate Impact	
	High Impact		Moderate Impact
	Moderate Impact		
	Moderate Barrier		
Moderate Barrier			Moderate Barrier
		Moderate Barrier	
		Moderate Barrier	
	Staff training and upskilling		
Effective		Neutral	
	Effective		
Neutral	Staff training and upskilling		
	Important		Important
		Important	
	Important		
Important	Budget availability;Technical compatibility;Regulatory requirements;Staff		
readiness		Accuracy and time saving	
	Cost	Yes	
freq	23	79	
35		87	39

40

53

25

41

40

42

42

51

42

39

40

41

48

39

43

25

36

36

40

41

64

38

50

31

42

38

31

2

3

3

